UNIT

ONE

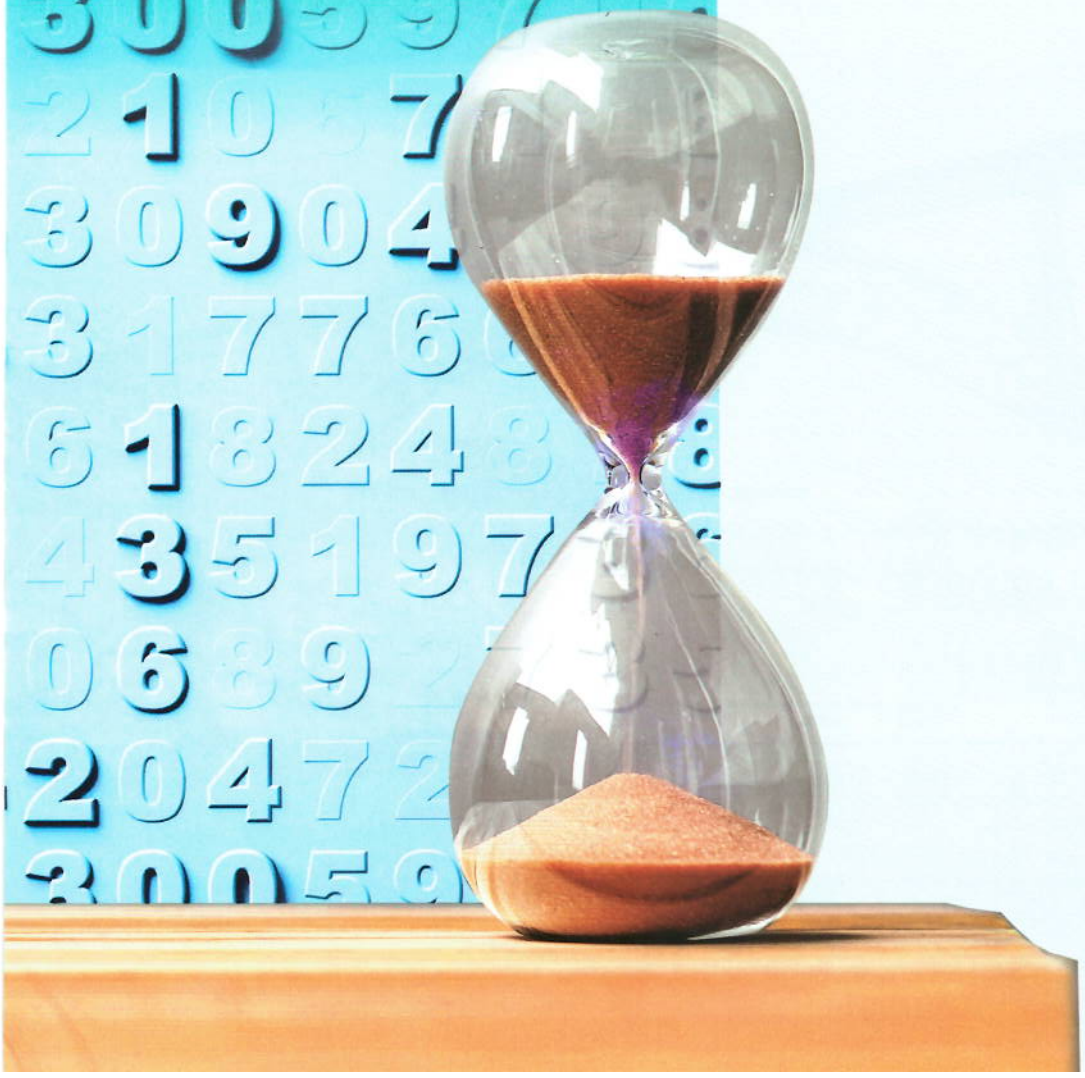
Differentiation and its applications

Exercise 1 Implicit and parametric differentiation.

Exercise 2 Higher derivatives of the function.

Exercise 3 Derivatives of exponential and logarithmic functions.

Exercise 4 Related time rates.



Exercise 1

On implicit and parametric differentiation

From the school book | Understand | Apply | Higher Order Thinking Skills

First Problems on implicit differentiation

1 Find $\frac{dy}{dx}$ for each of the following :

(1) $x^2 - 4y^2 + 7 = 0$

(2) $x^4 + 3y^4 - 2 = 0$

(3) $(x-3)^2 + (y+2)^2 = 25$

(4) $x^3 + y^2 - 7x + 5y = 8$

(5) $x^3 + y^3 - 2x + 3y^2 = 8$

(6) *(2nd Session 2009)* $x^2 + 9y^2 = 6xy + 4$

(7) $x^3 - 5xy + y^3 = 4x$

(8) $xy = 100(x + y)$

(9) $x^2 y^2 = 10$

(10) $x^2 y + y^2 x = 25$

(11) $x^n y^n = a^n$ where a is constant , n is positive integer.

(12) $x^n + y^n = a^n$ where a is constant , n is positive integer.

2 Find $\frac{dy}{dx}$ for each of the following :

(1) $\frac{2}{x} + \frac{3}{y} = 5$

(2) $\frac{x^3 + y^3}{x^3 - y^3} = 7$

(3) $\sqrt{x} + \sqrt{y} = 4$

(4) $y + \sqrt{xy} = 2x^2$

(5) $x \sin y = y$

(6) $\sin 2y = y \cos 2x$

(7) $3y = \sin 2x \cos y$

(8) $x \cos y + y \cos x = 1$

(9) $x^2 \sin y - y^2 \sin x = 9$

(10) $x \csc y = y \cot x$

(11) $\tan 2x + \cot y = xy$

(12) $\sin^2 x + 2 \cos y + xy = 0$

(13) $y \sec x + \tan x + x^2 y = 0$

(14) $a \cos bx + b \sin ay = c$ where a , b , c are constants.

3 **Find the slope of the tangent of the following curves at the indicated points :**

(1) $x^2 - 3xy + y^2 = 11$ at the point $(2 , -1)$ « $\frac{7}{8}$ »

(2) $xy^3 = 3$ at the point $(3 , 1)$ « $\frac{-1}{9}$ »

(3) $x^4 - 2x^3y^2 + xy^3 = 1$ at the point $(1 , 2)$ « 3 »

(4) $x^2 - xy + y^2 = 27$ at the point $(6 , 3)$ « undefined »

(5) $x\sqrt{y} + y\sqrt{x} = 6$ at the point $(1 , 4)$ « $\frac{-16}{5}$ »

(6) $\cos\sqrt{\pi y} = 3x + 1$ at the point $\left(\frac{-1}{3} , \frac{\pi}{4}\right)$ « -3 »

(7) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where a , b are constants at the point $(a\cos\theta , b\sin\theta)$ « $\frac{-b}{a}\cot\theta$ »

4 **Find the measure of the positive angle which the tangent to the given curve makes with the positive direction of the x-axis at the indicated points :**

(1) $x^3 - x^2y - x - 3y = 0$ at the point $(1 , 0)$ « $26° 34\tilde{}$ »

(2) $\frac{x}{y} - \frac{4y}{x} = 3$ at the point $(1 , -1)$ « $135°$ »

(3) $\sin 2x = \tan y$ at the point $\left(\frac{3\pi}{4} , \frac{3\pi}{4}\right)$ « zero »

Second Problems on parametric differentiation

5 **Find $\frac{dy}{dx}$ for each of the following curves at the given points :**

(1) 🕮 $x = 5t + 3$, $y = 16t^2 + 9$ at $t = 5$ « 32 »

(2) $x = at^2$, $y = 2at^6$ at $t = 1$ « 6 »

(3) $x = 4t$, $y = \frac{4}{t}$ at $t = 3$ « $\frac{-1}{9}$ »

(4) 🕮 $x = (t + 7)(t - 2)$, $y = (t^2 + 1)(t - 2)$ at $t = 1$ « zero »

(5) 🕮 $x = \sqrt{3t - 2}$, $y = \sqrt{4t + 1}$ at $t = 2$ « $\frac{8}{9}$ »

6 **Find $\frac{dy}{dx}$ for each of the following curves at the given points :**

(1) 🕮 $x = 3\cos 2\theta$, $y = 4\sin 3\theta$ at $\theta = \frac{\pi}{4}$ « $\sqrt{2}$ »

(2) 🕮 $x = \sin 2\pi\theta$, $y = \cos 2\pi\theta$ at $\theta = \frac{1}{6}$ « $-\sqrt{3}$ »

(3) $x = 5\cos^2\theta$, $y = 4\sin^2\theta$ at $\theta = \frac{\pi}{6}$ « $\frac{-4}{5}$ »

(4) $x = a\cos^3\theta$, $y = a\sin^3\theta$ at $\theta = \frac{\pi}{4}$ « -1 »

(5) $x = 5 + \sec^2 3\theta$, $y = 1 - \tan 3\theta$ at $\theta = \frac{\pi}{4}$ « $\frac{1}{2}$ »

(6) $x = \sec^2 \theta - 1$, $y = \tan \theta$ at $\theta = \frac{-3\pi}{4}$ « $\frac{1}{2}$ »

(7) $x = a \sec \theta$, $y = b \tan \theta$ at $\theta = \frac{3\pi}{4}$ « $\frac{\sqrt{2}\, b}{a}$ »

(8) $x = \cos \theta + \cos 2\theta$, $y = \sin \theta + \sin 2\theta$ at $\theta = \frac{\pi}{3}$ « $\frac{\sqrt{3}}{9}$ »

(9) $x = 3 \cos \theta - 2 \cos^3 \theta$, $y = 3 \sin \theta - 2 \sin^3 \theta$ at $\theta = \frac{\pi}{2}$ « zero »

(10) $x = a (\cos \theta + \theta \sin \theta)$, $y = a (\sin \theta - \theta \cos \theta)$ at $\theta = \frac{\pi}{6}$ « $\frac{1}{\sqrt{3}}$ »

7 **By using parametric differentiation, find :**

(1) Derivative of $(4x^3 - 9x^2 + 5)$ with respect to $(3x^2 + 7)$

(2) Rate of change of $(x + 3)(x - 2)$ with respect to $\frac{x-3}{x+2}$

(3) Derivative of $\frac{x+1}{x-1}$ with respect to $\sqrt{2x+1}$ at $x = 4$ « $\frac{-2}{3}$ »

(4) Derivative of $(x - \sin x)$ with respect to $(1 - \cos x)$ at $x = \frac{\pi}{3}$ « $\frac{\sqrt{3}}{3}$ »

Third Miscellaneous problems

8 **If f , g and h are differentiable functions with respect to x , complete the following using the values given in the opposite table :**

(1) $h(x) = 3 f(x) - 2 g(x)$, then $\dot{h}(1) =$ .....

(2) $h(x) = f(x) [5 + g(x)]$, then $\dot{h}(2) =$ .....

(3) $h(x) = f(x) \div [g(x) + 2]$, then $\dot{h}(1) =$ .....

(4) $h(x) = f[g(x)]$, then $\dot{h}(1) =$ .....

(5) $h(x) = g[3x - f(x)]$, then $\dot{h}(2) =$ .....

(6) $h(x) = [x^3 + g(x)]^{-2}$, then $\dot{h}(1) =$ .....

x	1	2
$f(x)$	– 1	4
$g(x)$	2	1
$\dot{f}(x)$	1	5
$\dot{g}(x)$	2	– 3

9 **Choose the correct answer from those given :**

(1) If $f(x) = 2a^6 + 5b^4$ where a , b are constants , then $\dot{f}(x) =$ .....

(a) $12a^5 + 20b^3$ (b) $6a^5 + 4b^3$ (c) 1 (d) zero

(2) If $h(x) = |x|$, then $\grave{h}(-5) =$ .....

(a) 5 (b) -5 (c) 1 (d) -1

(3) If $y = \sqrt{g(x)}$ and $\grave{g}(2) = 4$, $g(2) = 9$, then $\frac{dy}{dx} =$ ..... at $x = 2$

(a) $\frac{4}{3}$ (b) $\frac{2}{9}$ (c) $\frac{1}{6}$ (d) $\frac{2}{3}$

(4) If $x^3 y^2 = 1$, then $\left[\frac{dy}{dx}\right]_{y=1} =$ .....

(a) $-\frac{2}{3}$ (b) $-\frac{3}{2}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

(5) If $f(5x) = x^2 + x$, then $\grave{f}(2) =$ .....

(a) 5 (b) 1 (c) $\frac{3}{25}$ (d) $\frac{9}{25}$

(6) If $f(x) = 3x + g(x)$ and $\grave{g}(2) = -5$, then the slope of the tangent to the curve of the function f at $x = 2$ equals .....

(a) -2 (b) 3 (c) -15 (d) $\frac{1}{2}$

(7) If $xy = 1$, then $\frac{dy}{dx}$ equals each of the following except .....

(a) x^{-2} (b) $\frac{-1}{x^2}$ (c) $\frac{-y}{x}$ (d) $-y^2$

(8) If $\frac{x}{y} + \frac{y}{x} = a$ where a is constant , then $\frac{dy}{dx} =$ .....

(a) $\frac{x}{y}$ (b) $\frac{y}{x}$ (c) $\frac{ax}{y}$ (d) $\frac{ay}{x}$

(9) If $y = \frac{z+1}{z-1}$, $x = \frac{z-1}{z+1}$, then $\frac{dy}{dx} =$ ..... at $x = 2$

(a) $\frac{-1}{8}$ (b) $\frac{-1}{4}$ (c) $\frac{1}{4}$ (d) -4

(10) ***(Speciman 2025)*** The derivative of $(x^2 + 3)$ with respect to $\sqrt{x^2 + 16}$ at $x = 3$ equals .....

(a) 6 (b) 8 (c) 10 (d) 15

(11) If $x = 2t^2 + 3$, $y = \sqrt{t^3}$, then $\frac{dy}{dx}$ at $t = 1$ equals .....

(a) $\frac{1}{8}$ (b) $\frac{1}{4}$ (c) $\frac{3}{8}$ (d) $\frac{8}{3}$

(12) The rate of change of $\sin x^3$ with respect to $\cos x^3$ equals .....

(a) $-\cot x^3$ (b) $\cot x^3$ (c) $\tan x^3$ (d) $-\tan x^3$

(13) A point (x , y) moves on a curve such that $\frac{dx}{dt} = t^2$, $\frac{dy}{dt} = 2t$, then the slope of the curve at $t = 1$ equals .....

(a) $\frac{1}{2}$ (b) 2 (c) 4 (d) $\frac{1}{4}$

(14) The slope of the tangent to the curve $y = t^2 + 2t$, $x = t^2 - 2t$ equals ..... at $t = 2$

(a) $\frac{1}{3}$ (b) $-\frac{1}{2}$ (c) 2 (d) 3

(15) The tangent to the curve $x = t^2 - 1$, $y = t^2 - t$ is parallel to the x-axis at $t =$ .....

(a) zero (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{1}{2}$ (d) $\frac{-1}{\sqrt{3}}$

(16) The normal to the circle $x^2 + y^2 = 12$ at any point on it passes through the point .....

(a) (2 , 2) (b) (1 , 1) (c) (0 , 0) (d) (– 2 , – 2)

(17) ***(1st Session 2023)*** If the straight line $y = m x - a$ is the tangent to the curve $x^3 + y^2 = 5$ at the point (1 , 2) , then $m =$ .....

(a) $\frac{-3}{4}$ (b) $\frac{3}{4}$ (c) $\frac{4}{3}$ (d) $\frac{-4}{3}$

(18) The tangent to the curve whose parametric equations are $x = 3\cos\theta$, $y = 3\sin\theta$ where $(0 \le \theta \le \pi)$ is parallel to x-axis at $\theta =$ .....

(a) zero (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) π

(19) ***(Trial 2021)*** If $y^2 = 8x$ where $y > 0$, then the point lies on this curve at which $\frac{dy}{dx} = \frac{dx}{dy}$ is .....

(a) (0 , 0) (b) $(1 , 2\sqrt{2})$ (c) $(\frac{1}{2} , 2)$ (d) (2 , 4)

(20) ***(Trial 2021)*** If $y = f(x)$ where $y = \sqrt[3]{n^2 + 7}$, $6n^2 x + n = 1$, then the perpendicular to the curve of the function at a point lies on it and its x-coordinate equals zero is parallel to .....

(a) x-axis (b) y-axis

(c) the straight line $y = x$ (d) the straight line $y = -x$

(21) ***(1st Session 2024)*** The equation of the normal to the curve :

$x = t^2$, $y = 6t - 5$ at $t = 1$ is .....

(a) $4y + x - 5 = 0$ (b) $3x + y - 4 = 0$

(c) $3y + x - 4 = 0$ (d) $4x + y - 5 = 0$

(22) If $x = a\left(t - \frac{1}{t}\right)$, $y = a\left(t + \frac{1}{t}\right)$, then $\frac{dy}{dx} =$ .....

(a) $\frac{-x}{y}$ (b) $\frac{x}{y}$ (c) $\frac{-y}{x}$ (d) $\frac{y}{x}$

(23) If $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$, then each of the following is true except .....

(a) $\frac{dx}{d\theta} = y$ (b) $\frac{dy}{dx} = \cot\frac{\theta}{2}$ (c) $\frac{dy}{d\theta} + \frac{dx}{d\theta} = 1$ (d) $y\frac{dy}{dx} = a\sin\theta$

(24) If $y = f(x^2)$ and $\frac{dy}{dx} = 2x^3 - 4x$, then $\dot{f}(4)$ = ...

(a) zero (b) 2 (c) 6 (d) 14

(25) *(Speciman 2025)* If $y = \sec x$, $z = \tan x$, then $\frac{dy}{dz}$ = ...

(a) $\cos^3 x$ (b) $\sin x$ (c) $\cos x$ (d) $\tan x \sec x$

(26) *(2nd Session 2022)* If $x = a\sec^3\theta$, $y = a\tan^3\theta$, then $\frac{dy}{dx}$ = ... at $\theta = \frac{\pi}{4}$ where the constant $a \neq 0$

(a) $\frac{\sqrt{2}}{2}$ (b) $\frac{1}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) zero

(27) *(1st Session 2022)* If $2xy + \pi\sin y = 2\pi$, then $\frac{dy}{dx}$ = ... at the point $\left(1 , \frac{\pi}{2}\right)$

(a) $\frac{-\pi}{2}$ (b) $\frac{\pi}{2}$ (c) π (d) $-\pi$

(28) If $f(x) = \frac{2}{x+1}$, $g(x) = 3x$, then $\frac{d}{dx}\left[(f \circ g)(x)\right]$ = ... at $x = -2$

(a) $\frac{-2}{25}$ (b) 6 (c) $\frac{1}{25}$ (d) $\frac{-6}{25}$

(29) If $f(x) = x^2$, $g(x) = \cot x$ and $h(x) = (f \circ g)(x)$, then $\dot{h}\left(\frac{\pi}{4}\right)$ = ...

(a) -4 (b) 4 (c) zero (d) -1

(30) Let $f(x) = 2x^3 + 1$, $h(x) = x^2$, then $\dot{f}(h(3))$ = ...

(a) zero (b) 9 (c) 486 (d) 2916

(31) If $f(x) = 3x^2 - 2$, then $(f \circ f)^{\cdot}(-1)$ = ...

(a) -36 (b) -18 (c) zero (d) 18

(32) Which of the following curves has a tangent with constant slope ?

(a) $x = \sin t$, $y = \cos t$ (b) $x = t^2$, $y = 3t^2$

(c) $x = 2t - 1$, $y = t^2 - 4$ (d) $xy = 7$

(33) If $f(x) + \dot{f}(x) = x^3 + 5x^2 + x + 2$ where $f(x)$ is a polynomial , then $f(x)$ = ...

(a) $x^3 + 2x^2 - 3x$ (b) $3x^2 - 4x - 3$

(c) $x^3 + 2x^2 + 5$ (d) $x^3 + 2x^2 - 3x + 5$

(34) If $f(2x+1) = x \cdot h(2x-5)$ and $h(1) = 2$, $\grave{h}(1) = 4$, then $\grave{f}(7) =$ .....

(a) 4 (b) 7 (c) 11 (d) 13

(35) *(Trial 2021)* If $\grave{f}(x) = \frac{1}{1-x^2}$, $g(x) = \tan x$, then $(f \circ g)\grave{}(x) =$ .....

(a) $\sec^2 x$ (b) $\sec^2 x \tan^2 x$ (c) $\cos^2 x$ (d) 1

(36) If $x = \tan y \csc y$, then $\frac{dy}{dx} =$ .....

(a) csc y cot y (b) csc y – cos y (c) csc y – sin y (d) sin y + sec y

(37) If $x = \tan y$ where y is an acute angle , then $\frac{dy}{dx} =$ .....

(a) $x^2 - 1$ (b) $\frac{1}{x^2-1}$ (c) $\frac{1}{x^2+1}$ (d) $\frac{-1}{x^2+1}$

(38) *(2nd Session 2021)* If $x = \tan(1+y)$, then $\frac{dy}{dx} =$ ..... at $x = 2$

(a) 0.5 (b) 0.2 (c) 1 (d) 5

(39) *(Speciman 2025)* If $\grave{f}(x) = \sqrt{2x^2-1}$, $y = f(x^2)$ then when $x = 1$, $\frac{dy}{dx} =$ .....

(a) 1 (b) 2 (c) – 2 (d) – 1

(40) The tangent to the curve $x^2 - xy + y^2 = 27$ at the point (6 , 3) makes an angle of measure ..... with the positive direction of the x-axis.

(a) 90° (b) zero (c) 45° (d) 60°

(41) If $y^2 = 2 + xy$, then $\frac{dy}{dx}$ ..... at $x = 1$

(a) has a unique value (b) has two values

(c) has an infinite number of values (d) undefined

(42) The rate of change of the volume of a sphere with respect to its area when its radius is 2 cm. equals .....

(a) $\frac{1}{2}$ cm^2 (b) 1 cm. (c) $\frac{1}{2}$ cm. (d) 1 cm^2

(43) *(1st Session 2021)* If $y = \frac{1}{(x+1)^2}$, then the rate of change of $\frac{1}{y}$ with respect to x^2 at $x = 1$ is .....

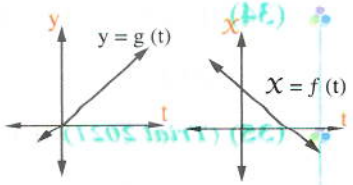
(a) $\frac{1}{2}$ (b) 4 (c) 2 (d) $\frac{1}{4}$

(44) *(Trial 2021)* If $\sin x \cos y = \frac{1}{2}$ where x , y are measure of acute angles , then $\frac{dy}{dx} =$ ..... at $x = \frac{\pi}{4}$

(a) 1 (b) – 1 (c) zero (d) $\frac{1}{2}$

(45) If the two opposite figures show the relations :

$y = f(t)$, $x = g(t)$, then $\frac{dy}{dx}$ could be equals .....



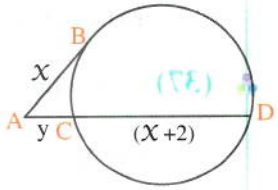
(a) zero (b) a positive number.

(c) a negative number. (d) a non-negative number.

(46) In the opposite figure :

If $\overline{AB}$ is a tangent to the circle

, then $\frac{dy}{dx}$ = ... at $x = 4$

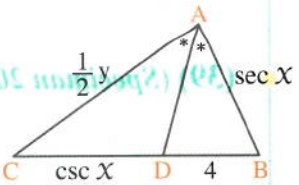


(a) 0.4 (b) 0.6 (c) 0.8 (d) 0.9

(47) In the opposite figure :

If $\overrightarrow{AD}$ bisects $\angle BAC$

, then $\frac{dy}{dx}$ = ...



(a) $-\csc 2x \cot 2x$ (b) $-2 \csc x \cot x$

(c) $-2 \csc 2x$ (d) $-2 \csc 2x \cot 2x$

10 If $y = (x^3 + 2)^7$ **Prove that :** $\frac{dx}{dy} = \frac{x^3 + 2}{21 x^2 y}$

11 *(2nd Session 2003)* If $x = (y^3 - 1)^5$ **Prove that :** $\frac{dy}{dx} = \frac{y^3 - 1}{15 x y^2}$

12 If $\frac{2}{\sqrt{x} + \sqrt{y}} = 5$ **Prove that :** $\left(\frac{dy}{dx}\right)^2 = \frac{y}{x}$

13 If $(y - a)^n - (x - a)^n = a^n$, where a is constant. **Prove that :** $\frac{dy}{dx} = \left(\frac{x - a}{y - a}\right)^{n-1}$

14 If $y^2 + 2ax - a^2 = 0$, where a is constant. **Prove that :** $\left(\frac{dy}{dx}\right)^2 + \frac{2x}{y} \times \frac{dy}{dx} = 1$

15 If $y^2 = x + \sqrt{1 + x^2}$ **Prove that :** $2\sqrt{1 + x^2}\,\frac{dy}{dx} = y$

16 If $y = \left(\frac{x^2 - 2}{x^2 + 2}\right)^n$ **Prove that :** $\frac{dx}{dy} = \frac{x^4 - 4}{8nxy}$

17 📖 **Find the value of the parameter z at which the curve :**

$x = 2z^3 - 5z^2 - 4z + 12$, $y = 2z^2 + z - 4$

(1) Has a horizontal tangent. **(2)** Has a vertical tangent. « $-\frac{1}{4}$, $-\frac{1}{3}$ or 2 »

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18 If $x = (1 - \sqrt[3]{y})(1 + \sqrt[3]{y} + \sqrt[3]{y^2})(1 + y + y^2)$ **Prove that :** $\frac{dy}{dx} = \frac{-1}{3y^2}$

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19 If $\sin x \cos y - \sin y \cos x = 1$ **Find** $\frac{dy}{dx}$ where : $x , y \in]0 , 2\pi[$ « 1 »

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20 If $\sin^2 x + \cos^2 y = \frac{1}{4}$ **Prove that :** $\frac{dx}{dy} = \sin 2y \csc 2x$

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21 If $x = \cos(xy)$ **Prove that :** $\frac{dy}{dx} + \frac{1 + y \sin(xy)}{x \sin(xy)} =$ zero.

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22 Prove that the derivative of the even function is odd and the derivative of the odd function is even.

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23 *(2nd Session 2011) , (1st Session 2004)* Find the points on the curve : $x^2 + xy + y^2 = 3$ at which the tangent is parallel to the y-axis. « (2 , –1) , (–2 , 1) »

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24 ***(2nd Session 2005)* Find the equation of the tangent to the curve :**

$x^2 + 2\sqrt{xy} + y^2 - 21 = 0$ at the point (1 , 4) which lies on it. « $8x + 17y - 76 = 0$ »

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25 ***(2nd Session 2012)* Find the equation of the normal to the curve :**

$x^3 + y^3 = 6xy$ at the point (3 , 3) which lies on it. « $x = y$ »

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26 ***(1st Session 2018)* Find the equations of the tangent and the normal to the curve :**

$x = \sec\theta$, $y = \tan\theta$ at $\theta = \frac{\pi}{6}$ « $3x + 6y - 4\sqrt{3} = 0$ »

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27 ***(1st Session 2012)* Find in terms of π the equation of the tangent to the curve :**

$y = \cos(x + y)$ whose slope $= \frac{-1}{2}$ where $0 \le x \le \pi$ « $2x + 4y - \pi = 0$ »

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28 ***(1st Session 2014)*** Find the equations of the two tangents to the curve $xy = 8$ which are parallel to the straight line : $y + 2x = 9$ « $y + 2x - 8 = 0$, $y + 2x + 8 = 0$ »

29 *(1st Session 2009)* Find the equations of the normal to the curve $x^2 + y^2 - x + 2y - 2 = 0$ at point of intersection with x-axis. « $3y = 2x - 4$, $3y = -2x - 2$ »

30 *(1st Session 2006)* Find the equation of the tangent to the curve $x^2 + y^2 = 50$ which makes an isosceles triangle with the two axes in the first quadrant. « $y + x - 10 = 0$ »

Higher skills

31 **Choose the correct answer from those given :**

(1) If $f(x) = g(x^2)$, then $\frac{\grave{f}(2)}{\grave{g}(4)} =$ .....

(a) 1 (b) 2 (c) 3 (d) 4

(2) If the function g is the inverse of the function f where f and g are differentiable functions on $\mathbb{R}$ and $\grave{f}(a) = 2$, $f(a) = b$, then $\grave{g}(b) =$ .....

(a) $\frac{1}{2}$ (b) 2 (c) $\frac{2}{3}$ (d) 1

(3) If $y = f(\tan x)$, $z = g(\sec x)$, where $\grave{f}(1) = 2$, $\grave{g}(\sqrt{2}) = 4$, then the rate of change of y with respect to z at $x = \frac{\pi}{4}$ equals .....

(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$ (c) 1 (d) zero

(4) If $x = \sqrt{\tan y + \sqrt{\tan y + \sqrt{\tan y + \cdots}}}$, then $\frac{dy}{dx} =$ .....

(a) $(2x - 1)\cos^2 y$ (b) $\cos^2 y$

(c) $(2x - 1)\sec^2 y$ (d) $(x^2 - x)\sec^2 y$

(5) If $f(x) = x^2$, then $(f \circ \grave{f})(1) =$ .....

(a) 2 (b) 4 (c) 8 (d) 9

(6) If $y = x^2 + 5x + 3$, then $\frac{dy}{d(x^2)} =$ ..... at $x = 1$

(a) $\frac{7}{2}$ (b) $\frac{5}{7}$ (c) 7 (d) $\frac{7}{5}$

(7) *(Trial 2021)* If $x = \sec y$ where $y \in]\frac{\pi}{2}, \pi[$, then $\frac{dx}{dy} =$ .....

(a) $x\sqrt{x^2 + 1}$ (b) $-x\sqrt{x^2 + 1}$ (c) $-x\sqrt{x^2 - 1}$ (d) $x\sqrt{x^2 - 1}$

(8) If $y = f(f(f(x))$ and $f(1) = 1$, $\grave{f}(1) = 3$ then $[\grave{y}]_{x=1} =$ .....

(a) 3 (b) 9 (c) 27 (d) 81

32 If $x = \sin y$, y is an acute angle. **Prove that :** $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

33 If $y = f(g(x))$, $f(x) = \frac{x}{\sqrt{x^2+1}}$, $g(x) = \frac{1}{x}$

Prove that : $\frac{dy}{dx} = \begin{cases} \frac{-1}{x^2\sqrt{1+x^2}} & \text{at } x > 0 \\ \frac{1}{x^2\sqrt{1+x^2}} & \text{at } x < 0 \end{cases}$

34 PQR is a triangle with area A , the point R moves on the straight line $y = 2x$ if P $(\ell , 0)$, Q $(0 , k)$ where ℓ , k are two (+ve) constant numbers

Prove that : $\frac{dA}{dx} = \begin{cases} -\ell - \frac{k}{2} & , x < \frac{k\ell}{k+2\ell} \\ \ell + \frac{k}{2} & , x > \frac{k\ell}{k+2\ell} \end{cases}$